

ISTRES AP, France

WMO: 076470

Lat: 43.523N

Lon: 4.922E

Elev: 25

StdP: 101.03

Time Zone: 1.00 (EUC)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-1.8	-0.3	-9.6	1.7	1.8	-7.1	2.1	2.9	19.4	6.8	17.4	6.7	5.2	350	0.557

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	33.2	20.7	31.9	20.2	30.6	20.0	23.4	28.6	22.6	27.8	21.8	27.1	5.5	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.0	16.7	25.7	21.0	15.7	25.1	20.1	14.8	24.5	69.8	29.0	66.6	27.8	63.8	27.1	27.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
16.5	14.3	12.4	-3.6	36.1	1.6	1.5	-4.8	37.2	-5.7	38.1	-6.6	39.0	-7.8	40.1		
			-4.7	24.6	1.9	0.8	-6.1	25.2	-7.2	25.7	-8.2	26.2	-9.6	26.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	15.6	7.3	7.8	11.1	14.0	17.9	22.5	24.8	24.4	20.6	16.6	11.1	8.1
	DBStd	6.89	3.12	3.30	2.84	2.60	2.70	2.92	2.28	2.21	2.61	3.21	3.50	3.17
	HDD10.0	294	94	72	21	2	0	0	0	0	0	2	28	75
	HDD18.3	1649	343	294	223	131	40	3	0	0	9	70	218	317
	CDD10.0	2323	10	11	56	122	245	373	460	446	317	207	61	16
	CDD18.3	636	0	0	0	1	26	127	202	187	76	17	0	0
	CDH23.3	5387	0	0	1	16	197	1103	1898	1638	481	53	0	0
	CDH26.7	1872	0	0	0	1	34	376	755	605	99	3	0	0
Wind	WSAvg	5.3	5.5	5.8	5.6	5.5	5.2	5.1	5.3	4.9	4.9	4.9	5.4	5.3
Precipitation	PrecAvg	608	54	36	34	57	45	31	18	31	91	88	76	47
	PrecMax	869	166	120	93	141	107	99	58	109	238	158	220	167
	PrecMin	372	1	0	0	3	2	1	1	6	14	37	17	0
	PrecStd	139	41	31	26	34	28	26	17	24	52	37	54	42
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.2	18.6	22.3	25.4	30.3	34.8	34.9	35.0	31.0	26.6	20.8	17.2
		MCWB	11.8	11.3	13.1	15.6	18.8	22.0	20.9	21.1	19.6	18.0	15.8	13.3
	2%	DB	15.1	16.1	20.0	23.1	27.2	32.5	33.2	32.9	29.1	24.8	19.2	15.8
		MCWB	11.2	10.5	12.3	14.7	17.3	20.3	20.7	20.8	19.2	18.3	15.4	12.5
	5%	DB	13.8	14.8	18.2	21.3	25.4	30.6	32.0	31.3	27.7	23.4	17.9	14.6
		MCWB	10.6	10.1	11.7	13.8	16.5	19.7	20.2	20.5	19.0	17.7	14.6	11.7
	10%	DB	12.6	13.4	16.7	19.8	23.9	28.8	30.7	30.0	26.2	22.0	16.7	13.4
		MCWB	9.7	9.5	11.1	13.0	16.1	19.2	20.0	20.1	18.4	17.1	13.7	10.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.1	12.7	14.6	17.3	20.5	23.5	24.6	24.8	22.6	20.7	17.5	14.5
		MCDB	15.6	15.6	18.8	21.9	25.7	30.9	28.2	29.5	26.1	23.5	19.5	16.1
	2%	WB	12.1	11.7	13.4	15.7	19.1	22.2	23.4	23.5	21.5	19.7	16.1	13.2
		MCDB	14.4	14.8	17.7	20.9	23.9	28.2	28.6	28.5	25.6	22.8	18.4	15.2
	5%	WB	11.0	10.9	12.7	14.7	18.0	21.2	22.4	22.8	20.7	18.8	15.1	12.2
		MCDB	13.4	13.9	16.7	19.6	23.3	27.3	28.0	27.7	24.8	22.0	17.5	14.1
	10%	WB	10.0	9.9	11.9	13.9	17.1	20.3	21.5	22.0	19.9	17.8	14.0	11.1
		MCDB	12.2	12.8	15.8	18.2	22.0	26.6	27.5	26.9	24.2	21.0	16.4	13.1
Mean Daily Temperature Range	5% DB	MDBR	7.9	8.9	9.9	10.0	10.4	10.9	10.9	10.9	10.2	9.0	8.0	7.9
		MCDBR	9.3	11.2	12.3	12.3	12.8	13.0	12.3	12.7	11.8	10.1	8.9	8.9
		MCWBR	6.3	6.9	6.4	5.9	5.5	4.9	4.5	4.8	5.1	5.1	5.8	6.2
	5% WB	MCDBR	8.3	10.2	10.6	11.1	11.7	11.8	10.7	10.8	10.2	8.5	7.9	7.7
		MCWBR	6.2	6.9	6.2	6.0	5.9	5.1	4.8	5.4	5.5	5.1	5.7	5.9
Clear-Sky Solar Irradiance	taub	0.328	0.346	0.383	0.400	0.410	0.410	0.403	0.401	0.390	0.390	0.353	0.324	
	taud	2.468	2.414	2.322	2.298	2.320	2.348	2.369	2.378	2.397	2.391	2.443	2.483	
	Ebn at Noon	800	845	852	865	864	863	864	854	836	781	758	769	
	Edh at Noon	73	91	113	125	125	122	119	114	104	93	75	66	
All-Sky Solar Radiation	RadAvg	1.78	2.76	4.10	5.37	6.48	7.46	7.44	6.37	4.84	3.06	1.97	1.55	
	RadStd	0.19	0.26	0.30	0.36	0.43	0.30	0.27	0.20	0.27	0.27	0.22	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	+0.35	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (33 neighbors)	+0.29	N/A	N/A	+0.65	N/A	N/A	N/A	N/A	N/A	+46

Nomenclature: See separate page